

Ride Cancellation Prediction

Case Study 66 | Binary Classification

B.Tech CSE 2024-28 | Semester V | Machine Learning

The Question and the Prediction Point

Binary Target Definition

The goal is to predict if a ride request will be recorded as **Cancelled** by the driver.

- **Positive Class (1):** Status is 'Cancelled'.
- **Negative Class (0):** Status is 'Trip Completed' or 'No Cars Available'.

Supply shortages ('No Cars Available') are treated as negative for this specific cancellation model to ensure a precise operational signal.

Exclusion Logic

Predictions are made strictly at the **moment of request**. The following fields are excluded to prevent data leakage:

- **Driver ID:** Not known before assignment; structurally missing for supply gaps.
- **Drop Timestamp:** Future information relative to the request time.

The model relies solely on request-time signals: pickup point, time, and date.

Dataset and Data Quality

Descriptive Statistics

6,745 Total Requests	18.7% Cancellation Rate
42.0% Completion Rate	39.3% Supply Gap Rate

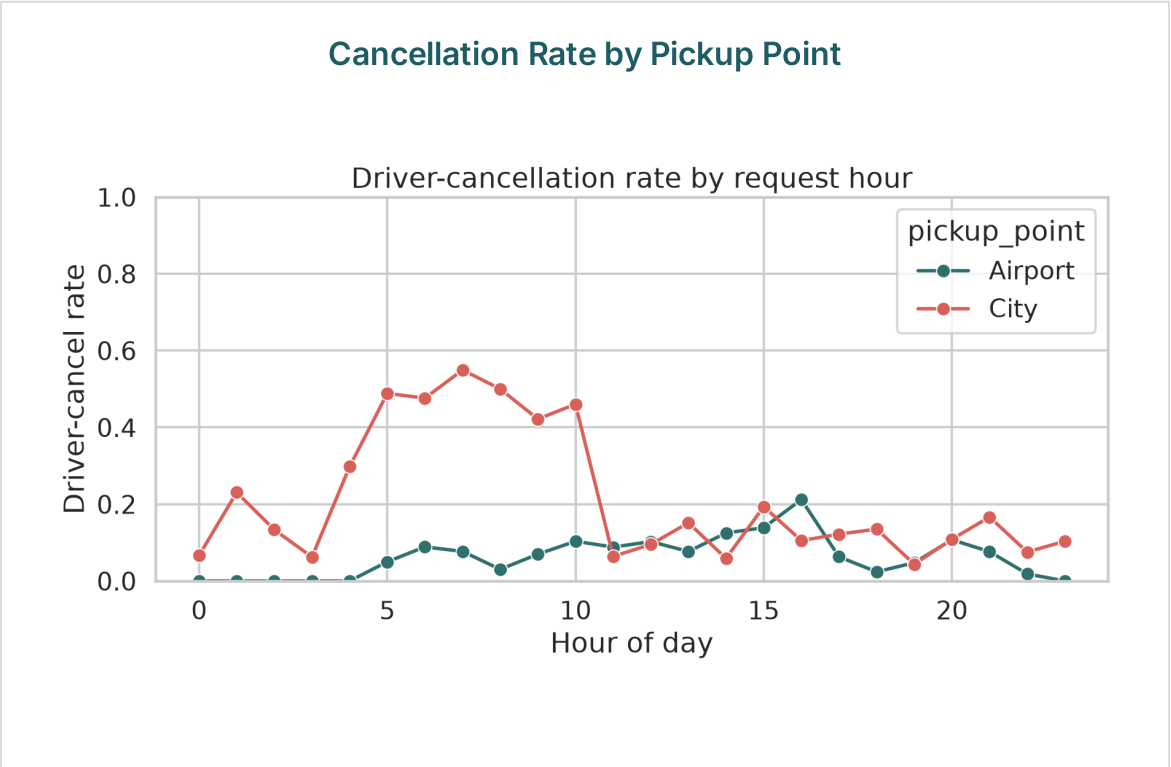
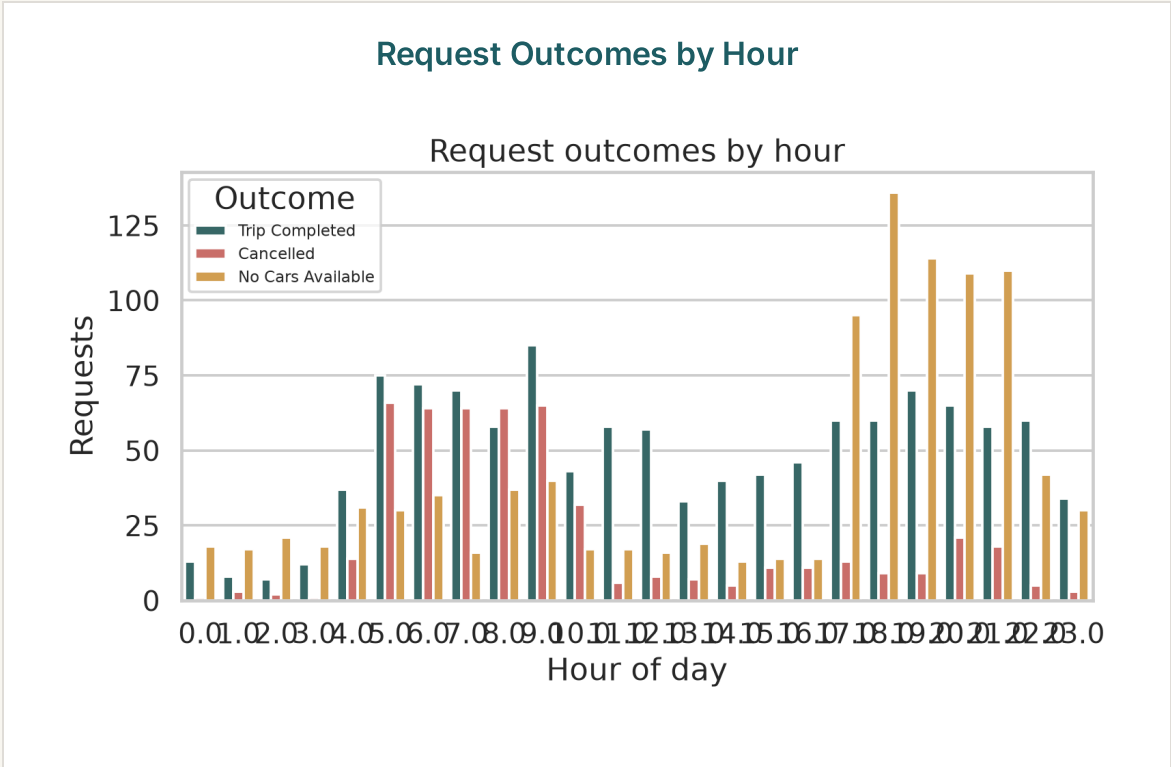
Source: Uber Request Data (July & Nov 2016). No duplicate rows found in the raw snapshot.

Structural Missingness

Field	Missing Values	Justification
Driver id	2,650	Matches 'No Cars Available' records.
Drop timestamp	3,914	Expected for all incomplete trips.
Request id	0	Complete identifier.
Pickup point	0	Categorical (Airport/City).

Structural missingness in post-dispatch fields confirms they cannot be used for request-time prediction.

What the Requests Look Like



Key Insight: Cancellations peak in the morning (City) and evening (Airport) hours. The binary target is strongly associated with the request time and location, justifying a request-time classifier.

From Raw Rows to Model Features

01

Normalisation

Lower-case snake_case headers;
datetime parsing for timestamps.

02

Feature Extraction

Deriving hour, day of week,
month, and weekend flag from
request time.

03

Transformation

One-hot encoding for categories;
median imputation and standard
scaling for numerics.

04

Splitting

75/25 stratified train/test split.
Balanced class weights applied at
training.

Final Model Features (observable at request time)

pickup_point

hour

day_of_week

month

is_weekend

Pipeline fits only on training data to prevent leakage.

Two Models, Same Inputs

Methods Compared

Model	Rationale
Logistic Regression	Transparent linear baseline for additive relationships.
Random Forest	Ensemble approach to capture non-linear interactions (e.g., Time × Location).

Both use class weighting to address the 18.7% target imbalance.

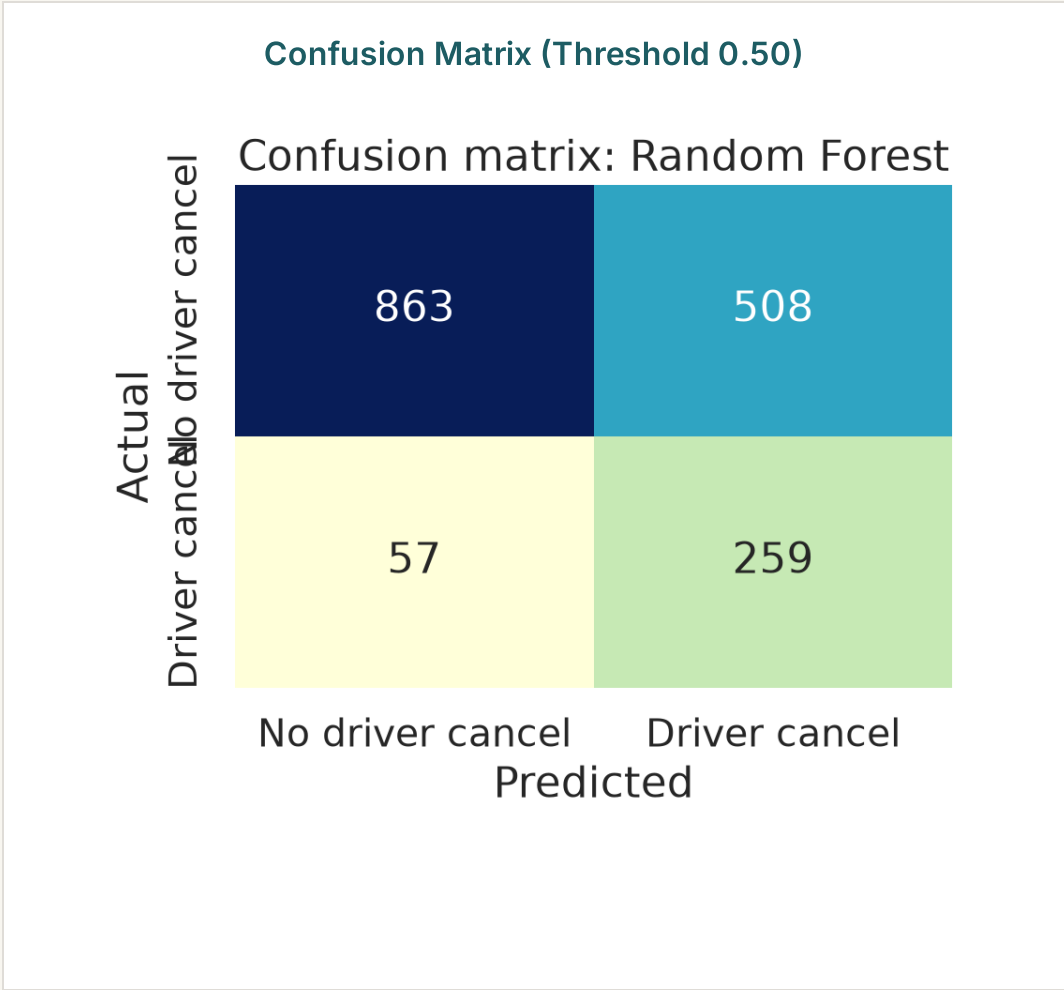
Selection Rule

The **Random Forest** was selected as the final model based on its higher Average Precision (AP) and ROC-AUC on the hold-out set.

Selection Criteria:

- **AP:** Robust measure for imbalanced ranking.
- **ROC-AUC:** Threshold-independent separation.
- **Recall:** Capability to flag actual cancellations.

Results on the Hold-out Set



Test Performance (n=1,687)

Metric	Random Forest (Final)	Logistic Regression
ROC-AUC	0.764	0.752
Avg. Precision	0.373	0.353
Recall	0.820	0.858
Precision	0.338	0.313
F1 Score	0.478	0.458

High recall indicates the model identifies 82% of actual cancellations, though precision is limited by false positives—a reasonable trade-off for early screening.

Streamlit Prototype

 **Ride Cancellation Prediction**

Pickup: Airport

Hour: 18:00

Day: Friday

Month: July

Estimate cancellation risk

Estimated probability

68.4%

Higher risk alert

Interactive demonstration in `app/streamlit_app.py`

Safe Use Policy: Risk scores support screening and resource planning; they must not be used for disciplinary actions.

What the Model Cannot Tell Us

Observational Data

The model identifies **associations**, not causes. Changing the request hour or location does not necessarily change the cancellation outcome.

Missing Operational Signals

Traffic, weather, surge pricing, driver proximity, and customer history are not in the source but likely drive real-world cancellations.

Supply Gaps are Separate

Requests that failed due to "No Cars Available" (39.3%) were treated as negative for this model. A separate supply-gap classifier is needed for full operational coverage.

Historical Bias

The 2016 dataset may not represent current market dynamics, vehicle supply, or rider behaviour.

Takeaways

The Model

A request-time Random Forest baseline identifies 82% of cancellations (Recall) with 33.8% Precision. It serves as a viable early-warning prototype.

The Trade-off

The score prioritises recall over precision. False alerts are common and must be managed through threshold selection and clear service policies.

The Deliverables

Full analysis pipeline, trained model, Streamlit app, 8-page final report, and 10-slide seminar presentation for Semester V submission.

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